

## Questions

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1. (a) (i) Given that  $f(x) = e^{-2x}$ , find  $f'(x)$  and  $f''(x)$ . [2]  
(ii) Hence, find the first three terms of the Maclaurin series for  $e^{-2x}$ . [2]

- (b) Hence, using a suitable value of  $x$ , show that

$$e^{-1/4} \approx \frac{25}{32}$$

[2]

- 2.** (i) Use the Maclaurin series for  $\sin x$  and  $\cos x$  to work out the series expansion of

$$\sin 2x(\cos x - \cos 3x)$$

up to and including the term in  $x^5$ .

[4]

- (ii) Hence show that, in exact surd form, an approximation to the least positive root of the equation

$$2 \sin 2x(\cos x - \cos 3x) = x$$

is

$$\sqrt{\frac{4 - \sqrt{10}}{12}}$$

[3]

- 3. (a)** Starting from the series given in the formulae booklet, show that the general term of the Maclaurin series for

$$2(1 - \cos x) - x \sin x$$

is

$$(-1)^r \frac{2(r-1)}{(2r)!} x^{2r} \quad [4]$$

- (b)** Show that

$$\lim_{x \rightarrow 0} \left[ \frac{2(1 - \cos x) - x \sin x}{(1 - \cos x)^2} \right] = \frac{1}{3} \quad [4]$$

4. Let  $f(x) = \ln(1 + \sin x)$ .

(a) (i) Determine  $f''(x)$ . [2]

(ii) Determine the first two non-zero terms of the Maclaurin expansion for  $f(x)$ . [3]

(iii) By considering the first two non-zero terms of the Maclaurin expansion for  $f(x)$ , find an approximation to

$$\int_0^{\frac{2}{5}} f(x) \, dx$$

Give your answer correct to 6 decimal places. [2]

5. (a) Given that

$$y = \cosh^{-1} \left( \frac{3+x}{2+x} \right)$$

show that

$$(x+2)\sqrt{5+2x} \frac{dy}{dx} + 1 = 0$$

[4]

(b) Hence find the first three terms in the Maclaurin series for  $\cosh^{-1} \left( \frac{3+x}{2+x} \right)$  in the form

$$a \ln \left( \frac{3+\sqrt{5}}{2} \right) + bx + cx^2$$

where  $a$ ,  $b$  and  $c$  are constants to be determined.

[5]

**6.**

$$y = (1 + \cosh x)^n \quad n \geq 2$$

**(a)** (i) Show that

$$\frac{d^2y}{dx^2} = n^2(1 + \cosh x)^n - n(2n - 1)(1 + \cosh x)^{n-1} \quad [4]$$

(ii) Determine an expression for

$$\frac{d^4y}{dx^4} \quad [2]$$

**(b)** Hence determine the first three non-zero terms of the Maclaurin series for  $y$ , giving each coefficient in simplest form. [2]

7. (i) Prove that, if  $y = \arctan x$ , then

$$\frac{dy}{dx} = \frac{1}{1+x^2} \quad [3]$$

- (ii) Find the Maclaurin series for  $\arctan x$ , up to and including the term in  $x^5$ . [4]

- (iii) Use the result of part (ii) and the Maclaurin series for  $\ln(1+t)$  to find the Maclaurin series for  $\ln(1+\arctan x)$ , up to and including the term in  $x^4$ . [5]

8. (a) The function  $f$  is defined as  $f(x) = \operatorname{sech}^2 x$ .

(i) Show that  $f^{(4)}(0) = 16$ .

[4]

(ii) Hence find the first three non-zero terms of the Maclaurin series for  $f(x) = \operatorname{sech}^2 x$ .

[2]

(b) Prove that

$$\lim_{x \rightarrow 0} \left( \frac{\operatorname{sech}^2 x - \cos(\sqrt{2}x)}{x^4} \right) = \frac{1}{2}$$

[4]

9. Using the Maclaurin series for  $\sin x$ , show that, for small values of  $x$ ,

$$\frac{x}{\sin x} \approx 1 + ax^2 + bx^4 + cx^6$$

where the values of  $a$ ,  $b$  and  $c$  are to be given in exact form.

[5]

10. (a) Use the Maclaurin series expansion for  $\ln(1+x)$  to show that the first three non-zero terms of the Maclaurin series expansion of

$$\ln\left(\frac{1+2x}{1-x}\right)$$

are

$$3x - \frac{3}{2}x^2 + 3x^3 \quad [2]$$

- (b) Sam attempts to use the series expansion found in part (a) to find an approximation for  $\ln 6$ .

Sam's incorrect working is shown below.

$$\begin{aligned} \text{Let } \frac{1+2x}{1-x} &= 6 \\ 1+2x &= 6-6x \\ 8x &= 5 \\ x &= \frac{5}{8} \end{aligned}$$

$$\begin{aligned} \text{So } \ln 6 &\approx 3\left(\frac{5}{8}\right) - \frac{3}{2}\left(\frac{5}{8}\right)^2 + 3\left(\frac{5}{8}\right)^3 \\ &\approx 2.02 \end{aligned}$$

Explain the error in Sam's working.

[2]

- (c) Use  $x = \frac{1}{4}$  in the series expansion found in part (a) to find an approximation for  $\ln 8$ .

Fully justify your answer.

[3]

**11. (a)** By using an appropriate Maclaurin series prove that if  $0 < x < 1$  then  $-\ln(1-x) > x$ . **[2]**

**(b)** Hence, by using a suitable substitution, deduce that  $\ln t > 1 - \frac{1}{t}$  for  $t > 1$ . **[1]**

**(c)** Using the inequality in part (b), and by making a suitable choice for  $t$ , determine which is greater,

$$\left(\frac{5}{4}\right)^5 \quad \text{or} \quad e$$

**[3]**

**12. (a)** Write down the first three non-zero terms of the Maclaurin series for  $\ln\left(\frac{1+2x}{1-x}\right)$ . [2]

**(b)** Use these three terms to show that

$$\ln\left(\frac{10}{7}\right) \approx \frac{183}{512} \quad [2]$$

**(c)** Dana uses the same first three terms of the series to approximate  $\ln 10$  and gets an answer of 2.67, correct to 3 significant figures. However,  $\ln 10 = 2.30$  correct to 3 significant figures.

Explain Dana's error. [2]

13. (a) Use the Maclaurin series expansions for  $\sin x$  and  $\cos x$  to determine the series expansion of

$$\left( \frac{\sin(x/2)}{x/2} \right) \cos\left(\frac{x}{3}\right)$$

in ascending powers of  $x$ , up to and including the term in  $x^4$ .

Give each term in simplest form.

[3]

- (b) Use the answer to part (a) and calculus to find an approximation, to 5 decimal places, for

$$\int_{\pi/6}^{2\pi/3} \left( \frac{1}{x} \left( \frac{\sin(x/2)}{x/2} \right) \cos\left(\frac{x}{3}\right) \right) dx$$

[3]

- (c) Use the integration function on your calculator to evaluate

$$\int_{\pi/6}^{2\pi/3} \left( \frac{1}{x} \left( \frac{\sin(x/2)}{x/2} \right) \cos\left(\frac{x}{3}\right) \right) dx$$

Give your answer to 5 decimal places.

[1]

- (d) Assuming that the calculator answer in part (c) is accurate to 5 decimal places, comment on the accuracy of the approximation found in part (b).

[1]

14. (a) Find and simplify the first five terms in the Maclaurin series for  $e^x$ . [2]

(b) Hence, or otherwise, write down the first five terms in the Maclaurin series for  $e^{-x}$ . [1]

(c) Use your answers, together with the identity  $\cosh x + 1 = 2 \cosh^2\left(\frac{x}{2}\right)$ , to show that the Maclaurin series for  $\cosh^2\left(\frac{x}{2}\right)$  is

$$a + bx^2 + cx^4 + \dots$$

where  $a$ ,  $b$  and  $c$  are rational numbers to be determined. [3]

- 15. (i)** Use the Maclaurin series for  $\ln(1+x)$  and  $\ln(1-x)$  to obtain the first three non-zero terms in the Maclaurin series for

$$\ln\left(\frac{(1+x)^2}{1-2x}\right)$$

State the range of validity of this series.

[4]

- (ii)** Find the value of  $x$  in the range of validity for which

$$\frac{(1+x)^2}{1-2x} = 2$$

Hence find an approximation to  $\ln 2$ , giving your answer to three decimal places.

[4]

**16.** Find the Maclaurin series for

$$e^{e^{x^2}-x}$$

up to and including the term in  $x^2$ .

[4]

- 17. (a)** Using the definitions of  $\sinh x$  and  $\cosh x$ , together with the Maclaurin series expansion of  $e^x$ , find the first three non-zero terms in the Maclaurin series expansion of  $x \cosh x - \sinh x$ . **[3]**
- (b)** Hence, by replacing  $x$  with  $ix$  in your answer to part (a), find the first three non-zero terms in the Maclaurin series expansion of  $x \cos x - \sin x$ . **[3]**